# Temporary Chat

02/17/2026 15:39:48

L11_S1_A - PDF You said Act as an expert lecture scribe and generate a strictly…

Thinking: Thinking: Analyzing the Project's Scope I've clarified the project's scope: …

> *L11_S1_A - PDF*

You said

Act as an expert lecture scribe and generate a strictly exam-focused, fully detailed lecture scribe in a complete, compilable LaTeX code block based exclusively on the attached PDF, ensuring no outside information is added. You must include every definition, notation, theorem, proof, and derivation exactly as presented in the slides—specifically covering the transformation of random variables ($Y=g(X)$) and functions of two random variables ($Z=X+Y$)—along with all illustrative examples and class problems found in the file. Format the document using the fancyhdr package with the left header containing 'Name: Nissarg Danidhariya \ Enrollment: AU2440261' and the right header containing 'Course: CSE400 \ Date: February 10, 2026'. To prevent header text overlapping, do not place the lecture title in the center header (\chead); instead, place the title "Lecture Scribe: Transformation of Random Variables" at the very top of the document body, centered and bolded, immediately after \begin{document}. Ensure the LaTeX code is strictly valid and free of compilation errors.

> *Thinking:*
>
> > *Thinking:*
> >
> > ***Analyzing the Project's Scope***

*I've clarified the project's scope: a LaTeX scribe derived from "L11_S1_A.pdf," exclusively focused on exam-relevant details. My confidence level is high on meeting the initial constraints. I am prioritizing a fully detailed, compilable LaTeX output. I will now start preparing to process the PDF and identify the pertinent information.*

***Confirming Key Requirements***

*I'm now completely clear on the project's scope. I can confirm the use of the attached PDF as the exclusive source. I must include all definitions, notations, theorems, proofs, derivations, and examples from the PDF. I'll need to include the transformation of both* $Y = g(X)$ *and* $Z = X + Y$ *(functions of two RVs). The headers will use* `fancyhdr`*. I am certain of all the header and title formatting requirements. The LaTeX output must be valid. My confidence remains high.*

***Deriving Transformations of RVs***

*I'm working on the derivation for* $Y = g(X)$ *, focusing on monotonically increasing and decreasing cases. I've worked through the CDF and PDF derivations, including the important step of inverting the function. The general formula now includes the absolute value of the derivative, capturing both cases. I've started the illustrative example using a uniform distribution and now I'm ready to find the PDF of the derived RV.*

***Calculating the PDF***

*I've just finished calculating the PDF of Y, given that* $Y = \sin\left(\frac{\pi x}{2}\right)$ *. I worked through the limits, and the derivative, and the PDF is* $\frac{1}{\pi\sqrt{1-y^2}}$ *for* $-1 < y < 1$ *. I have turned my focus to the transformation of two RVs:* $Z = X + Y$ *. I am considering the case of independent X and Y. I have two problems to work through.*

***Defining the LaTeX Structure***

*I have a concrete LaTeX structure now. I've designed sections for single and two random variable transformations, with an illustrative example and practice problems. My goal is to structure the information for clarity. Each derivation step will be numbered and boxed, so it's readily usable for the exam. I'm focusing on concise explanations for exam readiness.*

***Refining Source Citations***

*My initial citation strategy, considering direct inclusion within the LaTeX code, proves suboptimal. It could negatively impact compilation and readability. I've shifted to inline comments, referencing the source PDF slides directly, to maintain clean LaTeX while ensuring traceability. I am focusing on matching all content to the slides. The emphasis is on accuracy and a compilable output.*

***Implementing the Integral***

*I'm now implementing the integral for the sum of two random variables, focusing on Slide 7. It's the standard double integral, but the limits of integration are a bit tricky, and I'll need to confirm that my interpretation of those limits is accurate. I'm focusing on the integration region now for* $Z = X + Y$ *, using* $f_{XY}(x, y)$ *with the provided formula. I'm confident in capturing this key concept.*

***Defining the Scribe's Header***

*I've just finalized the document's header, incorporating student and course details. I have used a centered title for clarity and an outline section for easy navigation. I've designed the document's sections for single and two random variable transformations, including examples and problems.*

***Generalizing to Derivatives***

*I've just generalized the monotonically increasing/decreasing cases to a single formula. Considering that monotonicity gives a unique inverse, I have now included the absolute value of the derivative, ensuring the correct PDF regardless of increasing or decreasing behavior. I'm confirming all steps with careful derivation and now ready to proceed with another example.*

***Calculating the PDF***

*I've just finalized the PDF calculation for the illustrative example. My focus has been on finding the PDF of* $Y$ *when* $Y = \sin\left(\frac{\pi x}{2}\right)$ *, given* $X$ *follows a uniform distribution. The result,* $\frac{1}{\pi\sqrt{1-y^2}}$ *, confirms the solution's accuracy. I'm satisfied and now ready to document it. I will continue to organize the PDF example within my scribe.*

***Confirming Limits and Derivation***

> *I've just finalized the derivation, computing the PDF of Y. I've focused on ensuring the limits of $Y$ align with the example's domain. The calculation confirms the result of $\frac{1}{\pi\sqrt{1-y^2}}$ . I'm now proceeding with the case of two RVs: $Z = X + Y$ , where $X$ and $Y$ are independent. I will now derive the CDF and PDF of Z.*
>
> ***Verifying Content Alignment***
>
> *I'm now cross-referencing my work against the source PDF. Pages 1-2, showing the outline, are a match. I have verified that pages 3-4's derivation (CDF, PDF, and absolute value for the derivative) are accurate. The example on page 5 also aligns with my calculations and the resulting PDF. I'm now comparing the formulas for Z=X+Y.*

## Gemini said

Here is the compilable LaTeX code for the lecture scribe, strictly based on the provided PDF contents.

```
\documentclass[12pt]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath, amssymb, amsthm}
\usepackage{geometry}
\usepackage{fancyhdr}
\usepackage{graphicx}
\usepackage{float}

% Page geometry
\geometry{a4paper, margin=1in}

% Header and Footer setup
\pagestyle{fancy}
\fancyhf{} % Clear all header and footer fields
\lhead{Name: Nissarg Danidhariya \\ Enrollment: AU2440261}
\rhead{Course: CSE400 \\ Date: February 10, 2026}
\setlength{\headheight}{28pt} % Adjust for multiline header
```

```
\begin{document}

\begin{center}
    \textbf{\Large Lecture Scribe: Transformation of Random
Variables}
\end{center}

\section*{Outline}
\begin{enumerate}
    \item Transformation of Random Variables ($Y = g(X)$)
    \item Function of Two Random Variables ($Z = X + Y$)
    \item Illustrative Examples and Derivations
\end{enumerate}

\section{Transformation of a Single Random Variable}
\subsection*{Problem Definition}
Let $X$ be a random variable (RV) where the Probability Density
Function (PDF) $f_X(x)$ and Cumulative Distribution Function (CDF)
$F_X(x)$ are known a priori.
We define a new random variable $Y$ through the transformation:
$$ Y = g(X) $$
\textbf{Objective:} Find the PDF of the new random variable,
denoted as $f_Y(y)$.

\subsection*{Derivation: The CDF Technique}
To find $f_Y(y)$, we first determine the CDF $F_Y(y)$ and then
differentiate it with respect to $y$. We consider two cases based
on the monotonicity of $g(x)$.

\subsubsection*{Case 1: Monotonically Increasing Function}
If $g(x)$ is monotonically increasing, there exists a unique
inverse function $x = g^{-1}(y)$.
\begin{itemize}
    \item \textbf{Step 1: Determine the CDF of Y}
    $$ F_Y(y) = \Pr(Y \le y) = \Pr(g(X) \le y) $$
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    Since $g$ is increasing, the inequality direction is
preserved:
    $$ F_Y(y) = \Pr(X \le g^{-1}(y)) = F_X(g^{-1}(y)) $$

    \item \textbf{Step 2: Differentiate to find the PDF}
    $$ f_Y(y) = \frac{d}{dy} [F_Y(y)] = \frac{d}{dy} [F_X(g^{-1}
(y))] $$
    Using the chain rule:
    $$ f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) $$
    $$ f_Y(y) = f_X(x) \frac{dx}{dy} $$
\end{itemize}

\subsubsection*{Case 2: Monotonically Decreasing Function}
If $g(x)$ is monotonically decreasing:
\begin{itemize}
    \item \textbf{Step 1: Determine the CDF of Y}
    $$ F_Y(y) = \Pr(Y \le y) = \Pr(g(X) \le y) $$
    Since $g$ is decreasing, the inequality direction flips:
    $$ F_Y(y) = \Pr(X \ge g^{-1}(y)) = 1 - F_X(g^{-1}(y)) $$

    \item \textbf{Step 2: Differentiate to find the PDF}
    $$ f_Y(y) = \frac{d}{dy} [1 - F_X(g^{-1}(y))] = 0 - f_X(g^{-1}
(y)) \frac{d}{dy} g^{-1}(y) $$
    $$ f_Y(y) = - f_X(x) \frac{dx}{dy} $$
\end{itemize}

\subsubsection*{General Formula}
Since the PDF must always be non-negative, we combine both cases
using the absolute value:
$$ f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|
$$
Or equivalently:
$$ f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| \quad \bigg|_{x =
g^{-1}(y)} $$
```

```latex
\section{Illustrative Example}
\textbf{Problem:} Let $X$ be a random variable uniformly distributed over the interval $(-1, 1)$. Find the PDF of $Y$ given the transformation:
$$ Y = \sin\left(\frac{\pi X}{2}\right) $$

\textbf{Solution:}
\begin{enumerate}
    \item \textbf{Define $f_X(x)$:}
    $$ f_X(x) = \begin{cases} \frac{1}{2} & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases} $$

    \item \textbf{Find the Inverse Relation:}
    $$ y = \sin\left(\frac{\pi x}{2}\right) \implies x = \frac{2}{\pi} \sin^{-1}(y) $$

    \item \textbf{Compute the Derivative $\frac{dx}{dy}$:}
    $$ \frac{dx}{dy} = \frac{2}{\pi} \frac{d}{dy}[\sin^{-1}(y)] = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}} $$

    \item \textbf{Determine Limits for Y:}
    \begin{itemize}
        \item At $x = -1$: $y = \sin(-\frac{\pi}{2}) = -1$
        \item At $x = 1$: $y = \sin(\frac{\pi}{2}) = 1$
        \item Range: $-1 < y < 1$
    \end{itemize}

    \item \textbf{Calculate $f_Y(y)$:}
    $$ f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right| $$
    $$ f_Y(y) = \frac{1}{2} \cdot \left| \frac{2}{\pi \sqrt{1-y^2}} \right| $$
    $$ f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}, \quad -1 < y < 1 $$
\end{enumerate}

\section{Function of Two Random Variables}
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\textbf{Context:} We consider a derived random variable $Z$ which
is a function of two random variables $X$ and $Y$:
$$ Z = g(X, Y) $$
Specifically, we analyze the sum:
$$ Z = X + Y $$

\subsection*{Derivation of the CDF}
To find the distribution of $Z$, we start with the CDF
formulation:
$$ F_Z(z) = \Pr(Z \le z) = \Pr(X + Y \le z) $$

This probability is the double integral of the joint PDF $f_{XY}
(x, y)$ over the region where $x + y \le z$ (equivalently $y \le z
- x$).

\textbf{Integral Setup:}
$$ F_Z(z) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x}
f_{XY}(x, y) \, dy \, dx $$

\section{Class Problems (Exam Focus: 10 Marks)}
The following problems were presented for solution and proofs:

\begin{enumerate}
    \item \textbf{Sum of Independent Normal RVs:}
    Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$
be independent random variables.
    \textbf{Prove that:}
    $$ Z = X + Y \sim \mathcal{N}(0, 2) $$

    \item \textbf{Sum of Exponential RVs:}
    If $X$ and $Y$ are independent exponential random variables
with parameter $\lambda$, find the PDF $f_Z(z)$.

    \item \textbf{General Independence Case:}
    Find $f_Z(z)$ if $X$ and $Y$ are independent. (This generally
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leads to the convolution of the marginal PDFs).
\end{enumerate}

\end{document}
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